Pramaan — EPC Deviation Intelligence

The proof engine for construction documents · Spec-to-Site Deviation Sentinel (PS4) · ET AI Hackathon 2026 · Problem Statement 4

Repository: `github.com/bansalbhunesh/parth` · Judge Mode: `parth-tan.vercel.app/judge` · Evidence: `parth-tan.vercel.app/evidence` · API: `parth-1-ma30.onrender.com/health`

Phase-2 deadline (live Unstop record): **2026-07-22, 23:59:21 IST**

The wedge: Pramaan converts a spec/submittal mismatch into a cited commissioning failure mode and the schedule window to act — *before* the failed test becomes a project delay. The trace is owner requirement → deviation → commissioning test → lead-time-to-failure, with evidence at each step. It is a **benchmarked prototype**, not a field-validated product, and this document labels every number accordingly.

Evidence labels used throughout: **BENCHMARKED** from the frozen ps4_external_v1 (v1.2) run **DETERMINISTIC OFFLINE** reproducible with no API key **LIVE-MODEL** varies run-to-run **SYNTHETIC** authored corpus, true by construction **SCENARIO** modeled assumption, not validated **PENDING** not yet done.

1. Problem statement & PS4 alignment

In a hyperscale data-centre EPC build, the **design basis**, **vendor submittals**, and **governing standards** live in different systems and are reviewed by different people. A vendor deviation missed at submittal review resurfaces months later at commissioning — as a failed acceptance test, rework, and schedule slip. Found on submittal day it is a one-line RFI; found at commissioning it is a schedule event. PS4 asks for AI that reconciles specifications against submittals and standards; Pramaan does that and adds the commissioning/schedule consequence and the window to act.

Market context **SCENARIO**: India data-centre capex is large and growing (secondary research). These frame the opportunity; they are not Pramaan-specific results.

2. Product workflow

- Reconciles a design specification against a vendor submittal and the governing standards, and reports each genuine deviation with the standard it violates and a citation chain (required value, submitted value, source, confidence).
- For each deviation, predicts the **commissioning test** that will fail and the **lead-time-in-weeks** to fix it if caught now — the commissioning-risk twin — and connects it to schedule and supply-chain services.
- **Judge Mode (`/judge`)** is the 60-second flow: *Load deviation demo ★* (a natural-prose pair with a hidden 2N→N+1 and 10→8 min non-compliance) and *Load compliant demo ✓* (a fully compliant submittal — the correct answer is zero deviations). Users can upload their own spec + submittal (PDF / image / text).
- Every result shows a **truthful provenance chip** — live LLM, vision, deterministic rule floor, or OCR — and a zero-deviation result from the rule floor is labelled inconclusive, not a green pass.

3. Technical architecture (as actually executed)

Pramaan is **one compliance reasoning graph, a set of connected deterministic intelligence services, and a reliability layer** — *not* five co-operating LLM agents. Verify against `backend/orchestrator.py`. The runtime graph is:

`ingest → load_standards → validate → reconcile → retrieve → critique → cx_predict → format_output`

- **Exactly one node reasons with an LLM** — `reconcile` (extraction happens inside it; there is no separate extract node). `ingest`, `load_standards`, `validate`, `retrieve`, and `critique` are deterministic.
- Two **bounded cycles**: a retrieval tool-call (`retrieve` fetches a cited standard missing from context and loops back) and a self-critique / reflexion loop (`critique` drops equality-false-positives and duplicates, loops back on a failed check). Both are bounded by env vars; a sequential runner drives the same nodes if LangGraph is absent.
- `cx_predict` maps each deviation to its commissioning test via a rule table then a standards graph; it calls an LLM only as a **fallback** for equipment classes in neither.
- **Connected deterministic services** (separate endpoints, not graph nodes): schedule risk (graph CPM + beta-PERT Monte Carlo), supply-chain risk (closed-form ETA/risk), and the project/Cx graph. The **RFI Copilot is a service** (BM25 retrieval + an LLM only to phrase a cited answer), *not* a graph node.

4. Benchmark methodology & results

The headline evaluation is the frozen, provenance-tracked **ps4_external_v1 (v1.2)** benchmark: **53 spec–submittal pairs, 129 single-author-frozen labels, 17 datacenter system types, and 64 clean-negative (compliant) controls**. The featured configuration is `gemini-3.1-flash-lite` over a 3-pass repeated run. Source: `benchmarks/ps4_external_v1/reports/benchmark_card.json` (regenerate with `python scripts/benchmark_report.py`).

RECALL BENCHMARKED **0.862** mean semantic recall (range 0.841–0.873 across 3 passes).	PRECISION BENCHMARKED **0.953** mean semantic precision.	F1 BENCHMARKED **0.905** mean semantic F1.
FALSE-ALERT RATE BENCHMARKED **0.000** on the 64 clean-negative controls.	LATENCY BENCHMARKED **~2.5 s** p50 per pair (gateway/model dependent).	RULE BASELINE DETERMINISTIC OFFLINE **0.111** recall (7/63), 0 false positives — a deliberately low floor.

The scorer performs **one-to-one (bipartite) matching** — each finding is credited to at most one label and vice-versa, so a broad finding cannot inflate recall, and unmatched findings count as false positives. Recall counts not-run (quota/outage) pairs as misses. The reasoning core clears the deterministic floor on the same frozen labels (0.862 vs 0.111).

Provenance & review status (the limitations that must travel with the numbers).

- Fixtures are **team-authored**; **10** are derived from public primary sources (**5** with a verified public URL). SOURCE FILES NOT STORED YET
- Labels are **single-author frozen**. A second human reviewer (reviewer-2) has **not** run: TWO-PERSON ADJUDICATION PENDING.
- An automated consistency audit checks each label against its own documents (123/129 consistent, 6 flagged). This is **machine QA, not a human reviewer**, and does not substitute for the pending adjudication.
- This is a **benchmark result — not** a real-world-accuracy, field-validation, or customer claim.

A comparison model (`gemini-2.5-flash`) reached a higher peak recall (~0.95) but was slower and did not complete a clean 3-pass repeat; it is reported as an ablation, not the primary result. Full card and live status: `parth-tan.vercel.app/evidence`.

5. OCR, failover & reliability

PROVIDER FAILOVER **MEASURED**

gemini → Qwen gateway → Groq → Claude → local Ollama → deterministic rule floor. Only configured providers are tried. **Availability, not accuracy** — every leg is scored the same. Live chain at `/llm-check` (`?deep=1` for a reconcile-sized probe).

DETERMINISTIC FLOOR **DETERMINISTIC OFFLINE**

No silent zeros: if no provider answers, the rule engine returns headline deviations computed from the documents (low-recall by design, 0.111), never seeded labels. The UI labels a rule-floor 0-result as inconclusive.

OCR **MEASURED**

Reads scanned / image-only PDFs and uploaded images via Tesseract (text-first, best-effort). `render.yaml` builds from `Dockerfile.backend` (Tesseract in-image); `GET /ocr-check` reports live status. Hosts without Tesseract return a clear "OCR unavailable" message.

`/health` exposes the running commit and analysis mode; every endpoint returns 200 without a key (bundled ground-truth data), so a cold or key-less backend never fakes a live result.

6. Security & public-demo hardening (implemented)

Phase-5 shipped **demo-level** hardening (verified by `tests/test_security*.py`, `test_upload_hardening.py`, `test_prompt_injection.py`, `test_no_secrets.py`):

- **Optional token auth** on analysis endpoints and **per-IP rate limiting** (single-instance, in-memory).
- **Upload validation** — MIME + extension + magic-byte allowlist; rejects archives/executables/disguised files, oversized uploads (15 MB cap), and decompression-bomb images.
- **Prompt-injection-resistant** prompts and **no secret leakage** in status endpoints (booleans/caps only; the token and client IP are never returned).

Demo hardening, not production security. Rate limiting has no shared store, auth is an optional demo token (not access control / tenant isolation), and this is not a production-load story. These are named limitations, not claims. See `docs/SECURITY_DEMO_RUNBOOK.md`.

7. Standards, IP & provenance limitations

- Third-party standards content is a team-authored paraphrase from public/secondary references — proprietary standard text is not redistributed; product and standard names are trademarks of their owners, and Pramaan is not endorsed by them.
- Benchmark provenance is governed by `docs/CLAIMS_REGISTER.md` and `docs/SOURCE_PROVENANCE.md`: allowed vs banned wording, each quantitative claim paired with its limitation. "Derived from a public primary source" is not the same as a stored primary-source PDF, and we do not conflate them.
- No API keys are committed; credentials are external secrets. `THIRD_PARTY_NOTICES.md` records third-party artifacts.

8. Deployment & reproducibility

TESTS MEASURED

901 automated tests (API, agents, security, resilience, benchmark, failover) + 160 Playwright E2E journeys. Full CI: Python tests + lint + pip-audit + bandit, frontend types/tests/build/E2E, Docker.

REPRODUCE OFFLINE DETERMINISTIC OFFLINE

`python scripts/run_benchmark_suite.py --rule-only` (manifest + hash + rule baseline + report) — no API key, a judge's laptop is enough.

DEPLOY MEASURED

Frontend on Vercel, API on Render (Docker backend). `/health` shows the running commit; `verify_live.py` checks the deployment end-to-end.

Reproduce the benchmark card: `python scripts/benchmark_manifest_check.py`, `benchmark_hash_sources.py`, `benchmark_report.py`. Deterministic evals and unit tests run offline; live-model results are labelled and vary run-to-run.

9. Known limitations & roadmap

- **Benchmark:** team-authored fixtures, single-author frozen labels; source-file archiving and two-person (reviewer-2) adjudication are PENDING. Not field- or customer-validated.
- **Schedule risk is simulation**, not field-calibrated prediction; deviation probabilities are seeded, not empirical. Supply-chain intelligence is closed-form over static data (no live feed).
- **No measured coordination-hours study** yet; any cost/ROI figure is an explicitly-labelled scenario, not realized savings.
- **Prototype hardening:** single-instance in-memory rate limiting, optional demo token; synchronous LLM calls; not a production-load service.
- **Next milestones (trust-first):** archived source artifacts + expanded primary-source-derived pairs; reviewer-2 two-person adjudication and resolving the 6 audit-flagged labels; production hardening (shared-store auth/rate-limit, pgvector/Qdrant, async, delta re-checks).

10. Links

- **GitHub:** `github.com/bansalbhunesh/parth`
- **Live demo — Judge Mode:** `parth-tan.vercel.app/judge`
- **Evidence dashboard:** `parth-tan.vercel.app/evidence`
- **API health / OCR / LLM status:** `/health` · `/ocr-check` · `/llm-check`
- **Benchmark report + card:** `benchmarks/ps4_external_v1/reports/benchmark_report.md` · `benchmark_card.json`
- **Deck (PDF):** `docs/Pramaan_Deck.pdf` · **Claims register:** `docs/CLAIMS_REGISTER.md`
- **Demo video:** LINK PENDING — SEE DOCS/VIDEO_RUNBOOK.MD

Rubric mapping (PS4 weights: 25 / 25 / 20 / 15 / 15)

Criterion	How Pramaan addresses it	Honest status
Innovation (25%)	Spec → deviation → commissioning test → lead-time chain; a distinctive wedge beyond submittal review	Strong — the moat is the Cx-test / lead-time link, not module count
Business impact (25%)	Cost-of-late-nonconformance thesis; expected-value impact model (docs/BUSINESS.md) with low/base/high assumption bands, not a single deterministic number	SCENARIO base case ~3.2x on modeled assumptions, honestly negative in the low case; no paid-pilot validation yet
Technical excellence (20%)	LangGraph reasoning graph + one LLM core, deterministic services, 901 tests, CI, frozen benchmark	BENCHMARKED 0.862/0.953/0.905; reviewer-2 pending
Scalability (15%)	Modular, containerized; deterministic services run at high throughput; async job flow + input-hash cache; persisted case workflow (SQLite, tenant-isolated) for submittal→RFI→audit-log	Prototype: single-instance auth/rate-limit; LLM path is the scale constraint
User experience (15%)	Focused Judge Mode + streaming reasoning + evidence page; provenance chips	Lighthouse (/judge): Performance 53→92, Accessibility 92→100 this phase

Anticipated judge questions (honest answers)

Is the data real? The benchmark fixtures are team-authored — 10 derived from public primary sources, 5 with a verified URL — and source files are not stored in this benchmark yet. We do not call them real datasheets. The benchmark measures a reliable first-pass workflow, not real-world accuracy.

Is the benchmark independently reviewed? Not yet. Labels are single-author frozen; the automated consistency audit (123/129) is machine QA, not a second human reviewer. Two-person adjudication is the next validation milestone.

Why not "five agents"? Exactly one node reasons with an LLM (`reconcile`); ingestion, retrieval, self-critique, and commissioning mapping are deterministic and inspectable, plus deterministic schedule/supply/graph services. One reasoning graph + deterministic services, not five agents.

What if the LLM is down during judging? Every endpoint returns 200, the rule engine still finds headline deviations, and the UI labels that as the low-recall floor — not a clean bill of health. `/llm-check` reports the true status.

Evidence-safe proof points: "On the frozen ps4_external_v1 (v1.2) benchmark — 53 team-authored pairs, 129 single-author-frozen labels, 17 systems, 64 clean negatives — the featured 3-pass run reports recall 0.862, precision 0.953, F1 0.905, and 0 false alerts on the clean negatives, vs a deterministic rule baseline of 0.111." · "901 automated tests; full CI." · "Reviewer-2 human adjudication is pending; the automated cross-check is machine QA." · Every metric carries an evidence label. Results are observations at the audited commit, not evergreen badges.